

Lecture-9

The Application Layer

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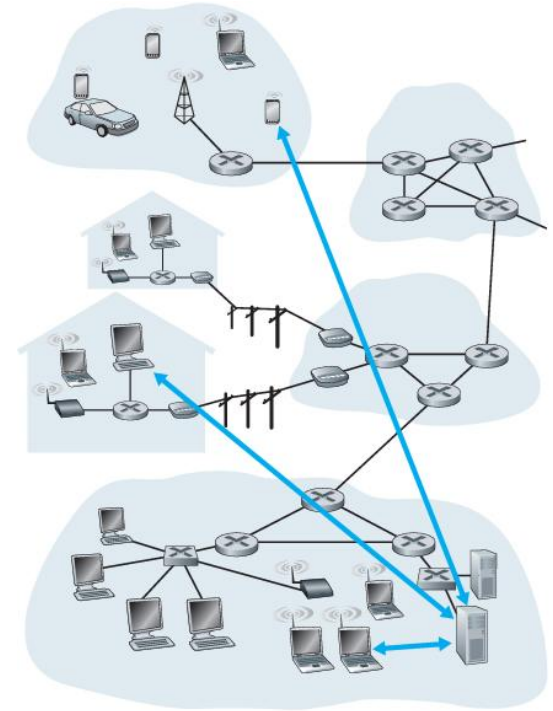
Network application Architecture

The two main **architectural paradigms** used in modern **network applications**:

- ✓ The client-server architecture
- ✓ The peer-to-peer (P2P) architecture

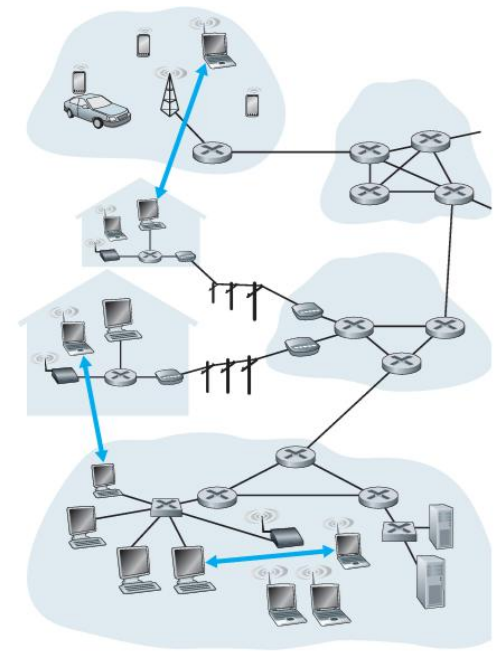
In a **client-server architecture**, there is an ***always-on*** host, called the server, which services requests from many other hosts, called clients.

Examples: Web, FTP, Telnet, and e-mail.



The client-server architecture

- ✓ In a **P2P architecture**, the application exploits (কাজে লাগান) direct communication between pairs of intermittently (মাঝে মাঝে অথবা অনিয়মিতভাবে) connected hosts, called peers.



P2P architecture

- ✓ The peers are not owned by the service provider, but are instead desktops and laptops controlled by users, with most of the peers residing in homes, universities, and offices.

- ✓ These applications include file P2P sharing protocol like, **BitTorrent protocol** and জনপ্রিয় চীনা ডাউনলোড ম্যানেজার **Xunlei**.
- ✓ ফাইলগুলো কোনো একটি মেইন সার্ভার থেকে না এসে, ছোট ছোট টুকরো হয়ে ব্যবহারকারীদের কম্পিউটার থেকে ডাউনলোড হয়, i.e. peer-assisted download acceleration.
- ✓ Internet Telephony (like Skype before 2012)

Seeding: Uploading

Leeching: Downloading

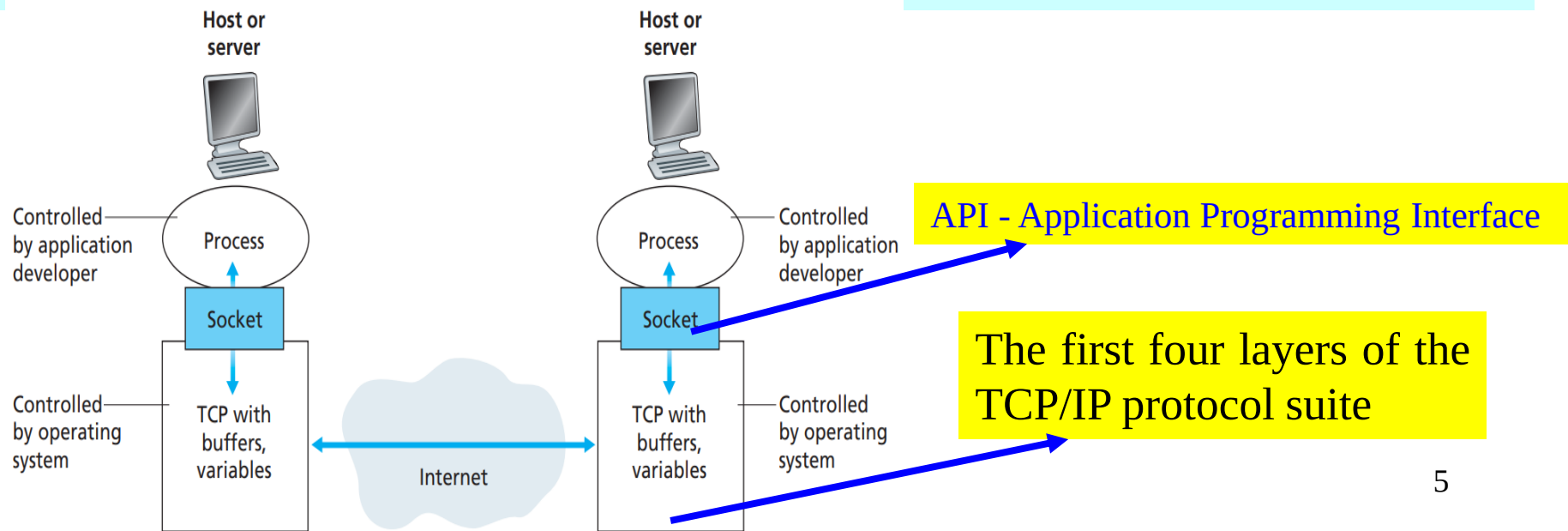
Q. Compare client-server and P2P architecture with example.

Application Programming Interface

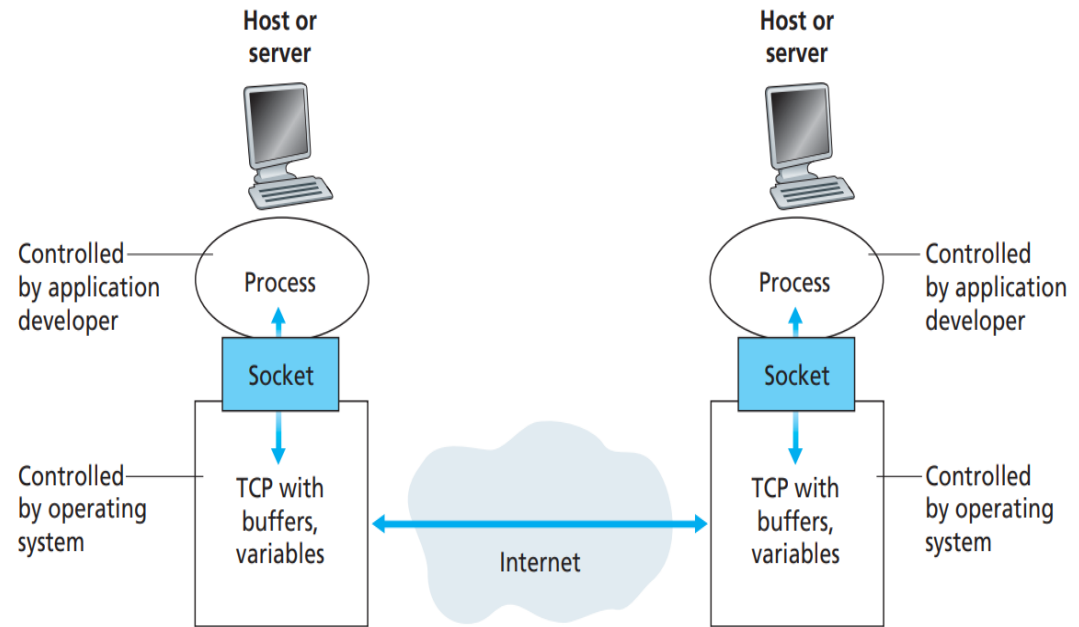
If we need a **process** to be able to communicate with another **process**, we need to consider three entities.

- (i) Process at the application layer
- (ii) the operating system that encapsulates the first four layers of the TCP/IP protocol suite
- (iii) API/ socket interface

A computer manufacturer needs to build the first four layers of the suite/stack in the operating system and include an API/ socket interface.



- ✓ Consider an analogy to help us understand processes and sockets. A process is analogous to a house and its socket is analogous to its door.



Q. Draw the arrangement of socket in process to process communication

- ✓ When a process wants to send a message to another process on another host, it shoves (or push) the message out its door (socket). This sending process assumes that there is a **transportation infrastructure** from sending door to the door of the destination process.
- ✓ Once the message arrives at the destination host, the message passes through the receiving process's door (socket), and the receiving process then acts on the message.

Addressing Processes

- ✓ A process running on **sending host** needs the full address of the process running on the **receiving host**. To identify each process, two pieces of information is needed : (1) the address of the host (IP address) and (2) the process identifier called port number
- ✓ This port number is needed because in general a host runs many network applications.
- ✓ For example, a Web server is identified by port number 80. A mail server process (using the SMTP protocol) is identified by port number 25.

Q. How a process is addressed in a host.

Application layer

- ✓ Users are familiar with several network applications: e-mail, World Wide Web, DNS, remote file access (Telnet service) and file transfer etc.
- ✓ Each application can be considered as a **set of programs** (or program) that run on different **end systems** and communicate with each other over the network.
- ✓ When a **new application is developed**, it is written for end machine but it does not need to be rewritten for networking devices, such as routers, that function at the network layer.

✓ This layer also contains **support protocols**, to allow the applications to function.

✓ An **application layer protocol** is associated with each **application**. Consider World Wide Web browsers that are available (Firefox, Safari, Internet Explorer etc.). You can use any one of these application programs to access a particular site on the Web is because they all use the same application layer protocol: HTTP (Hyper Text Transport Protocol).

Q. Give the idea of network application and application protocol.

This lecture will deal with open protocols:

- ✓ **Domain Name System (DNS)**
- ✓ **SMTP-The Simple Mail Transfer Protocol**
- ✓ **The World Wide Web**
- ✓ **URLs-Uniform Resource Locators**
- ✓ **Web Caching (Proxy Server)**
- ✓ **Telnet and Secure Shell**
- ✓ **File Transfer Protocol (FTP)**
- ✓ **Video streaming systems**
- ✓ **Network Management**

**Proprietary protocols:
Skype, Zoom**

Domain Name System (DNS)

- ✓ In Internet a host, web pages, mailboxes, web server and other resources are identified by **network addresses** (IP address) of the computers on which they are stored, these addresses are hard for people to remember.
- ✓ Initially the JU site were at a server of 128.111.24.41 and if the university moves the Web server to a **different machine with a different IP address**, everyone needs to be told the new IP address.
- ✓ Consequently, high-level, readable names were introduced like: ***www.juniv.edu/department/cse*** against the JU website regardless of its IP address.
- ✓ Nevertheless, the network itself understands only numerical addresses, so some mechanism is required to convert the **ASCII strings to network addresses**.

✓ DNS is a **tree type** global directory (**like distributed database**) that translates **machine** or **domain names** to **numerical IP address**.

✓ DNS is an application layer protocol, and every **Internet service provider (ISP)** has a **DNS server**. In the normal mode of operation, a hosts send UDP (connection less) queries to a DNS server. The DNS server either replies or directs the queries to another server.

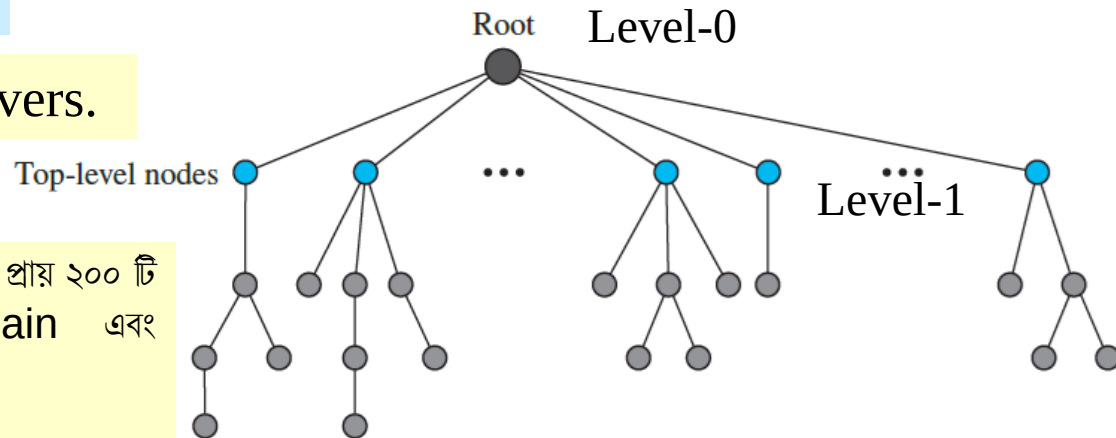
✓ A **distributed database** is a database in which storage devices are not all attached to a common processing unit such as the CPU, and which is controlled by a distributed database management system (together sometimes called a distributed database system). It may be stored in multiple computers, located in the same physical location; or may be dispersed over a network of interconnected computers.

Q. Why DNS server is used?

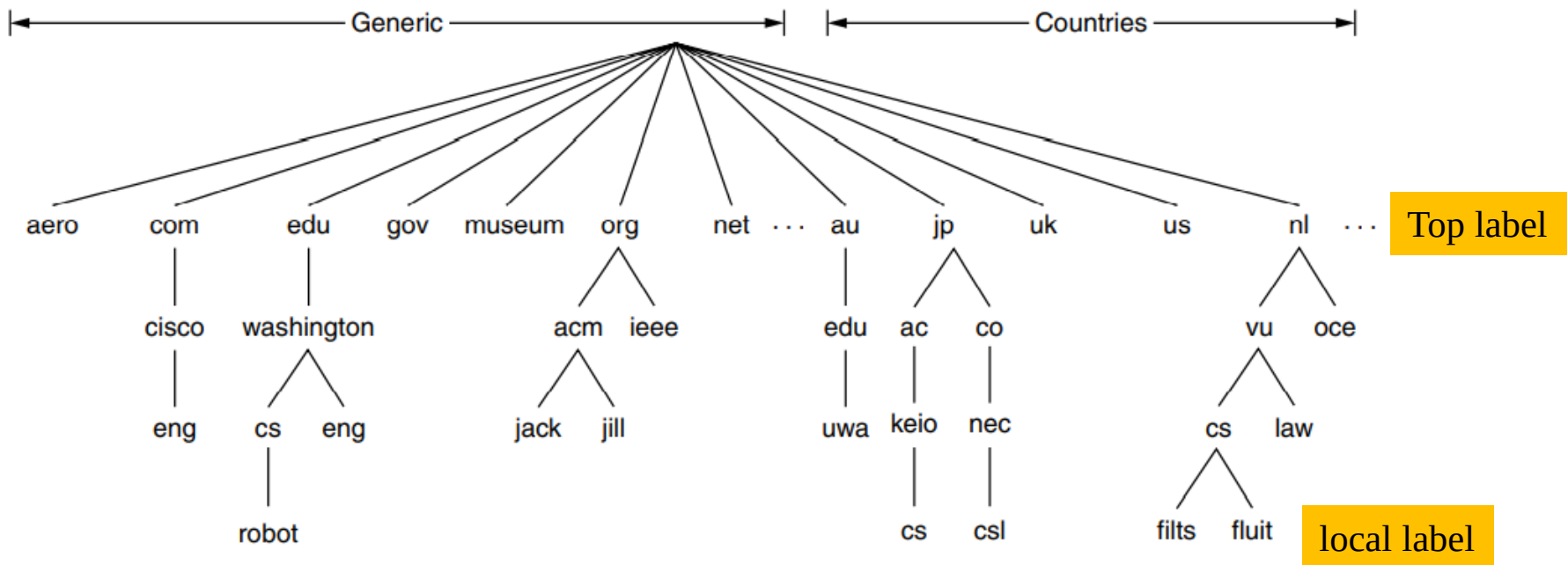
The DNS Name Space

- ✓ In theory at least, a single name server could contain the entire DNS database and respond to all queries about it. In practice, this server would be so overloaded as to be useless.
- ✓ To avoid above problem, the DNS name space is divided into **nonoverlapping zones**.
- ✓ Conceptually, the Internet is divided into over 250 top-level domains (first level of tree), where each domain covers many hosts. Each domain is partitioned into sub-domains, and these are further partitioned, and so on.

Portion of the hierarchy of DNS servers.



প্রতিটি Node এক একটি ডোমিন | Level-1- এ এরকম প্রায় ২০০ টি Node আছে যার প্রতিটি এক একটা Domain এবং প্রত্যেকটা ডমিনের আওতায় অজস্র URL আছে |



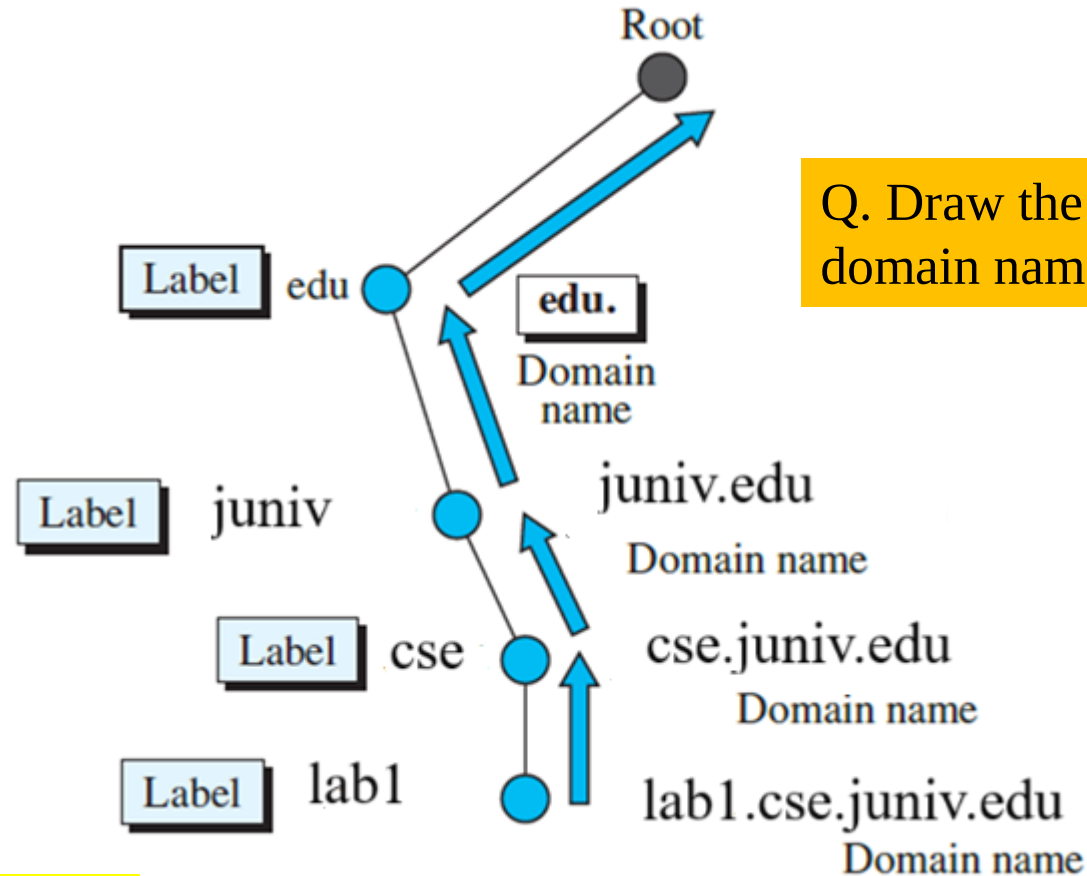
Tree type global directory

- ✓ The **top-level** domains come in two flavors: **generic** and **countries**.
- ✓ The few generic domains are: *com* (commercial), *edu* (educational institutions), *gov* (the U.S. Federal Government), *int* (certain international organizations), *mil* (the U.S. armed forces), *net* (network providers), and *org* (nonprofit organizations).
- ✓ The each node of the country domain assigns a country.

For the Internet, the top of the naming hierarchy is managed by an organization called **ICANN** (Internet Corporation for Assigned Names and Number).

Domain Name

Each node in the tree has a **label** and **domain name**. A full **domain name** is a sequence of labels separated by dots (.). The domain names are always read from

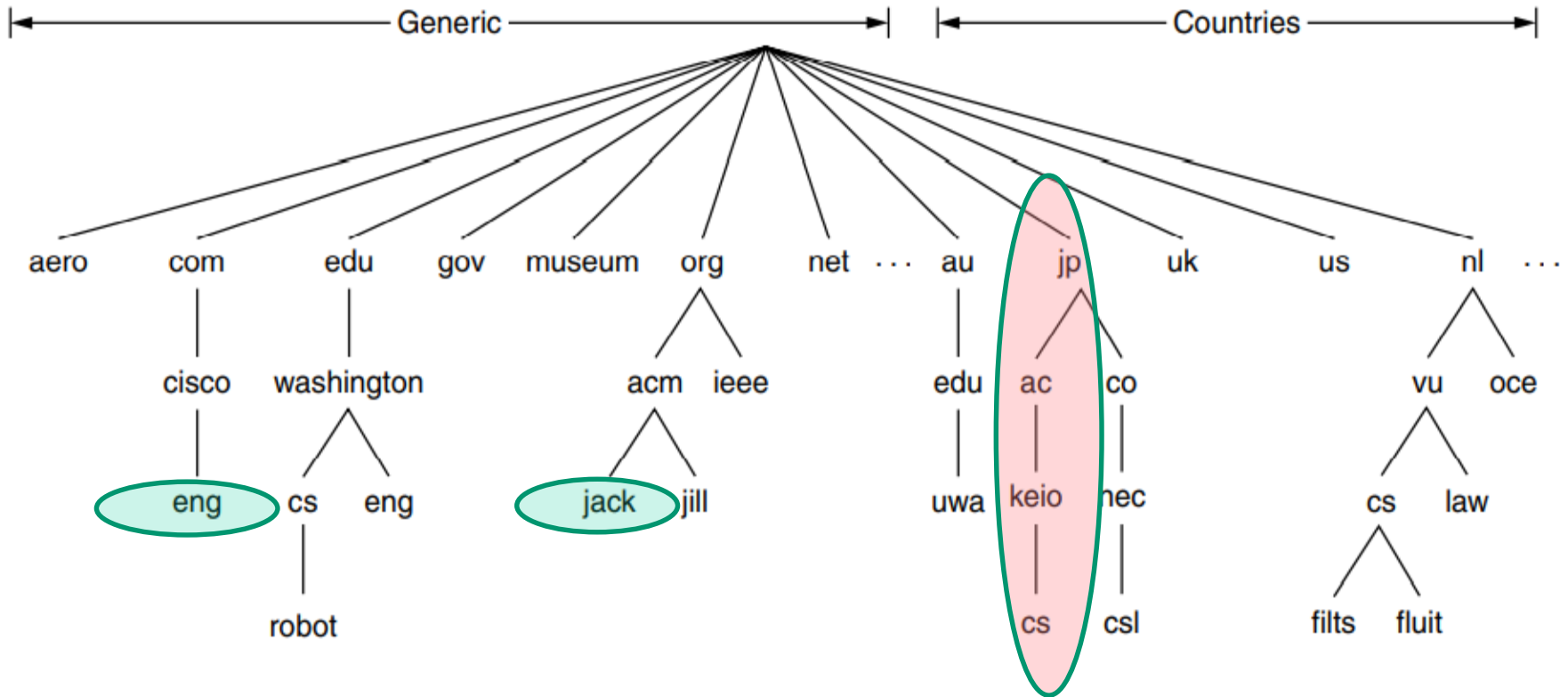


Q. Draw the hierarchy of domain name and labels.

The domain names are written with the local label first and the top domain last.

Domain names and labels

- ✓ A leaf domain may contain a single host, or it may represent a company and contain thousands of hosts.



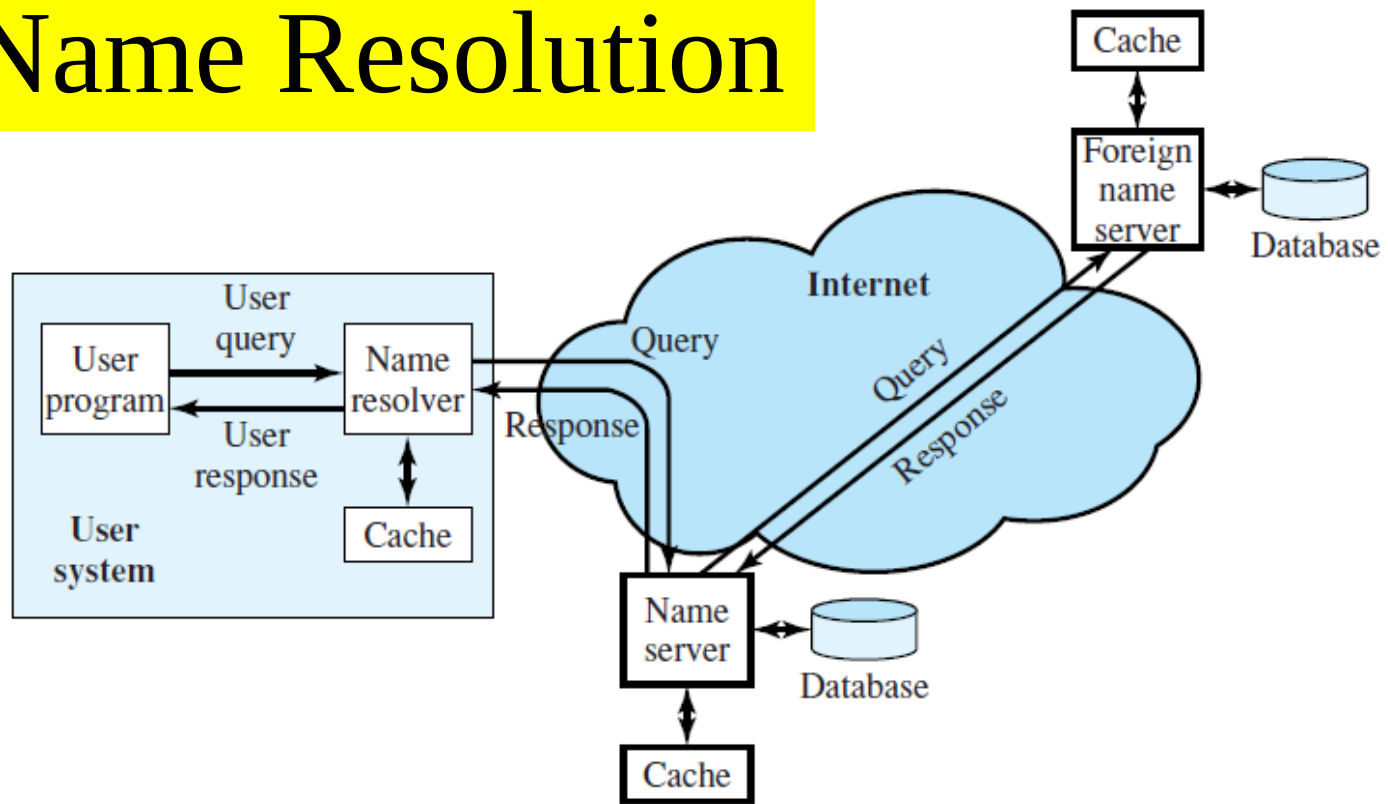
The domain names are written with the local label first and the top domain last.

cs.keio.ac.jp (CS department of Keio University, in Japan)

Fig.1 A portion of the Internet domain name space¹⁶

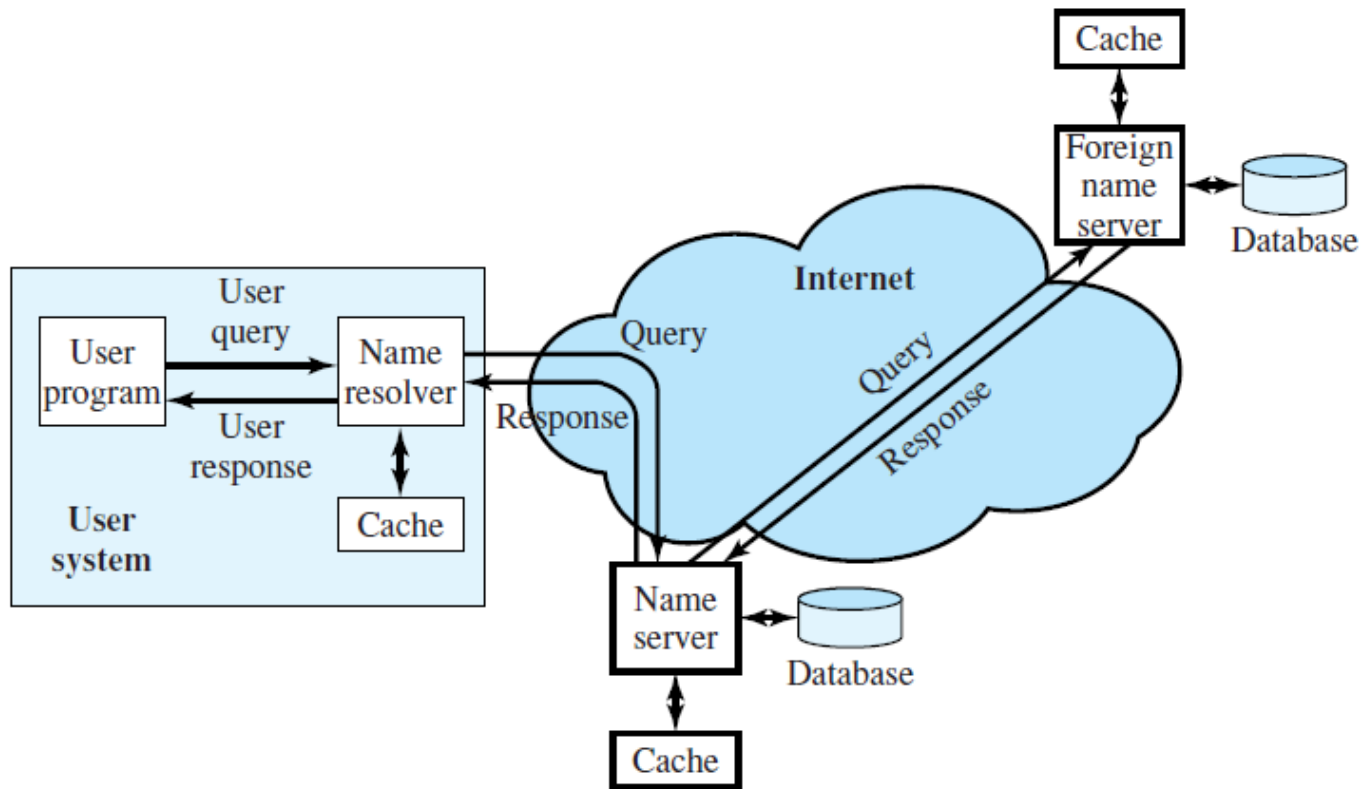
- ✓ Each domain controls the sub-domains under it.
- ✓ For example, Japan (*jp*) has domains *ac.jp* and *co.jp* that mirror *edu* and *com*. The Netherlands does not make this distinction and puts all organizations directly under *nl*.
- ✓ Thus, all three of the following are university computer science departments:
 1. *cs.keio.ac.jp* (Keio University, in Japan).
 2. *cs.vu.nl* (Vrije University, in The Netherlands).
 3. *cs.washington.edu* (University of Washington, in the U.S.).

DNS Name Resolution



DNS operation typically includes the following steps (Figure above):

1. A user program requests an IP address for a domain name.
2. The **local name resolver** (cache of browser) checks to see if the name is in its local cache, and, if so, returns the IP address to the requestor.
3. Otherwise, the name server queries other available name servers, starting down from the DNS tree or as high up the tree as possible.



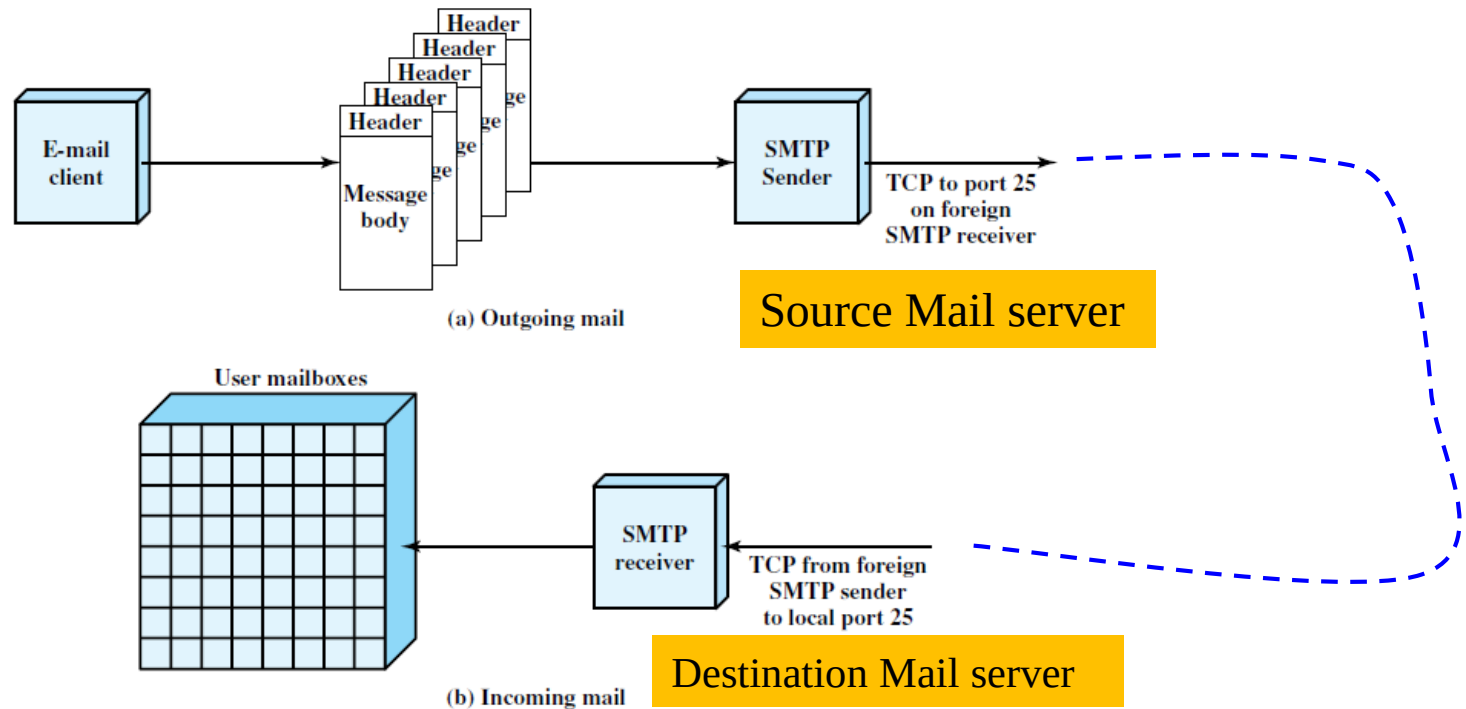
4. When a response is received at the **local name resolver**, it stores the name/address mapping in its local cache and may maintain this entry for the amount of time specified in the **time to live** field of the retrieved RR (*resource records*).

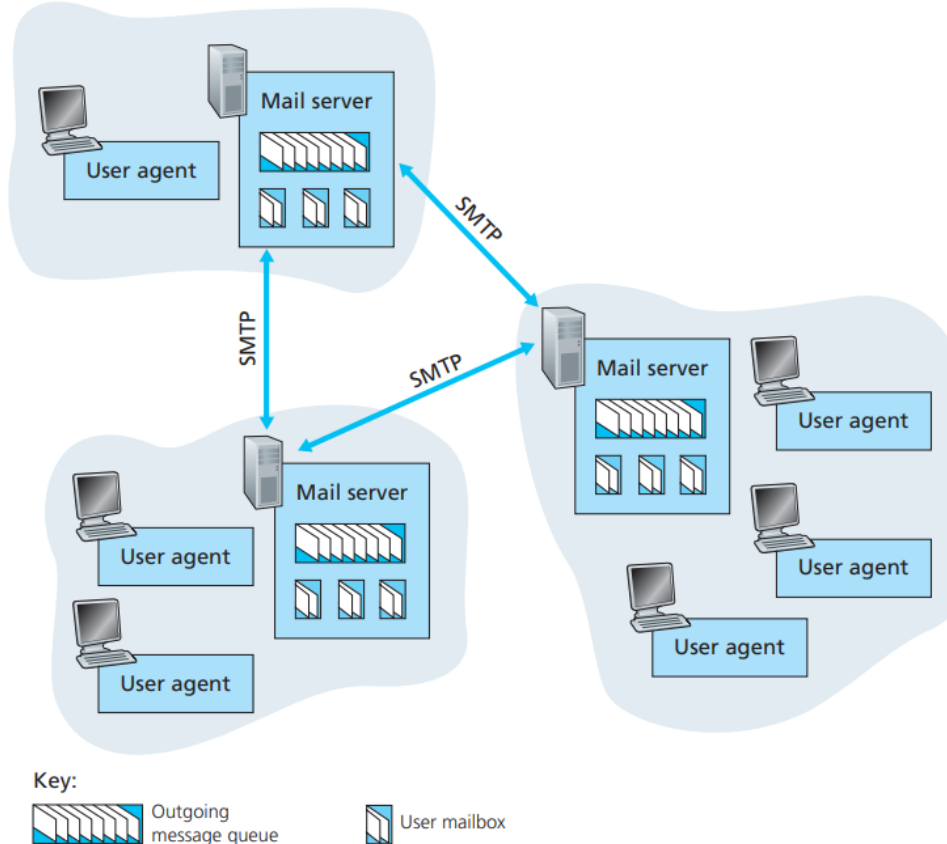
5. The user program is given the IP address or an error message.

Q. Give the steps DNS name resolution.

SMTP-The Simple Mail Transfer Protocol

✓The **Simple Mail Transfer Protocol (SMTP)** provides a basic electronic mail transport facility. This protocol (SMTP) transfers e-mail from the **mail server of a source** to the **mail servers of destinations** through TCP link.

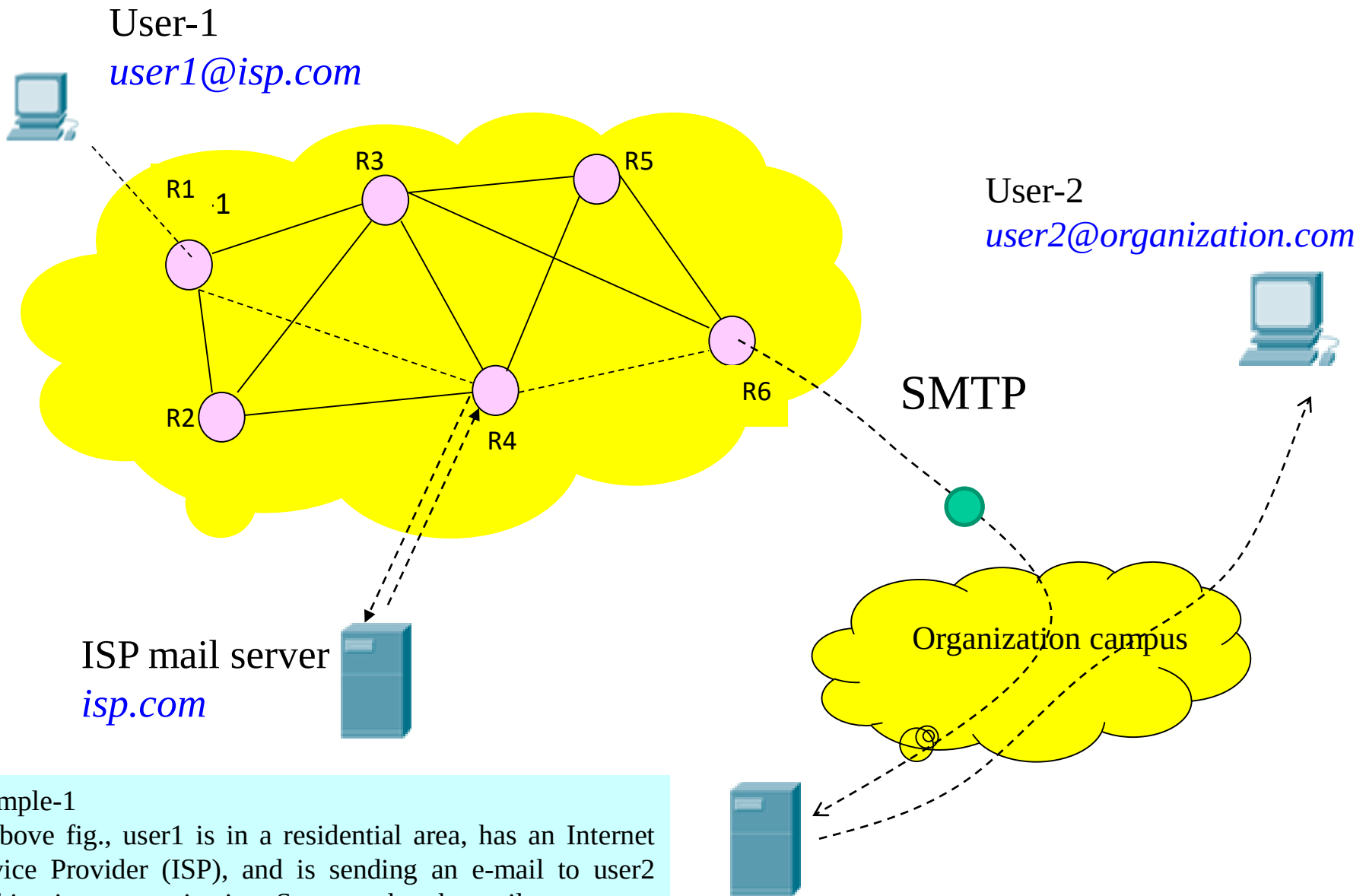




The three major components of e-mail system are:

- ✓ user agents,
- ✓ mail servers,
- ✓ Simple Mail Transfer Protocol (SMTP).

- ✓ Each user, has a **mailbox** located in one of the **mail servers**.
- ✓ A **mailbox** is a space in the **mail server** allocated to the user to keep its e-mail



Example-1

In above fig., user1 is in a residential area, has an Internet Service Provider (ISP), and is sending an e-mail to user2 working in an organization. Suppose that the mail servers are *isp.com* and *organization.com* respectively. The e-mail addresses of user1 and user2 are *user1@isp.com* and *user2@organization.com* respectively.

Organization mail server
organization.com

The procedure for e-mail exchange between user 1 and user 2 are as follows.

1. User1 provides user2's e-mail address (*user2@organization.com*) and composes its message.
2. User1 sends the message to its mail server (*isp.com*)
3. Server *isp.com* places the message in its queue.
4. SMTP on user1's mail server notices the message in the queue and opens a TCP connection with the organization mail server (*organization.com*)
5. Initial handshaking takes place between the two servers.
6. The message is sent to the organization mail server using the established TCP connection.
7. User2's mail server receives the message and then puts it in user 2's mailbox, ready to be retrieved by user2.

Post Office Protocol

- ✓ The most popular protocol used to transfer e-mail messages from a permanent **mailbox** to **local computer** is known as version 3 of the Post Office Protocol (POP3); a secure version of the protocol is known as POP3S.
- ✓ POP3 begins when the user agent (the client) opens a TCP connection to the mail server (the server) on **port 110**. The user first sends a login and password to authenticate the session.
- ✓ With the TCP connection established, POP3 progresses through three phases: authorization, transaction, and update.

- ✓ During the first phase, **authorization**, the user agent sends a username and a password (in the clear) to authenticate the user.
- ✓ During the second phase, **transaction**, the user agent retrieves messages; also during this phase, the user agent can mark messages for deletion, remove deletion marks, and obtain mail statistics.
- ✓ The third phase, **update**, occurs after the client has issued the quit command, ending the POP3 session; at this time, the mail server deletes the messages that were marked for deletion.

Q. Mention the function of three phases of POP3.

The World Wide Web

- ✓ Conceptually, *World Wide Web* (WWW) or simply *Web* consists of a vast, worldwide collection of contents in the form of **Web pages**, often called **pages** for short. The **Web pages** are accessible over the Internet.
- ✓ Each page may contain links to other pages anywhere in the world. The idea of having one page points to another called **hypertext**. A piece of text, icon, image and so on associated with another page is called a **hyperlink**.
- ✓ A Web page (also called a document) consists of objects. An object is simply a file—such as an HTML file, a JPEG image, or a video clip—that is addressable by a single URL.

✓ With a **web browser**, one can view **web pages** (**web page** may contain text, images, videos, and other multimedia) and navigate between them via hyperlinks.

URLs—Uniform Resource Locators

✓ If every page were somehow assigned a unique name, there would not be any ambiguity in identifying pages.

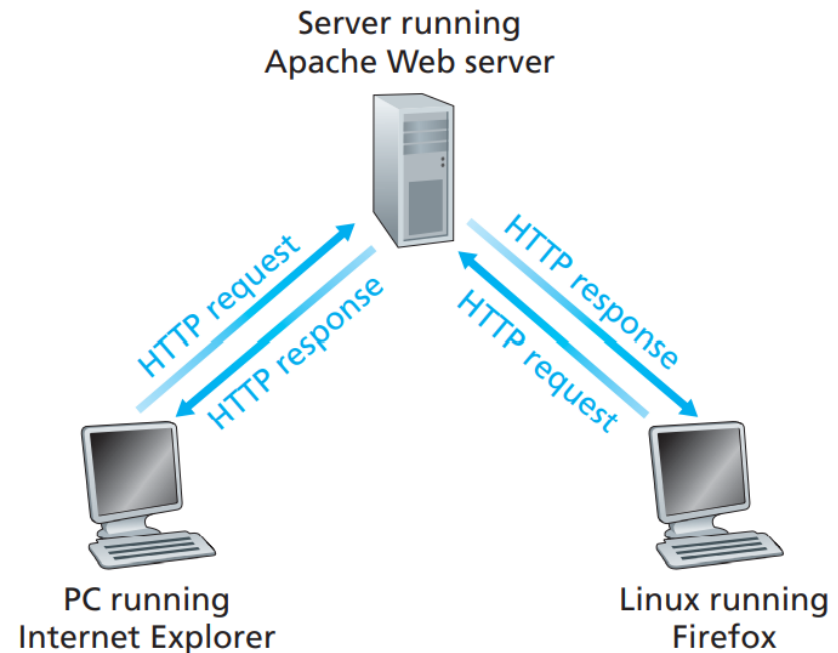
✓ URLs have three parts: the **protocol** (also known as the scheme), the **DNS name** of the machine on which the page is located, and a **local name** uniquely indicating the specific page (usually just a file name on the machine where it resides). As an example, the Web site for the author's department contains several videos about the university and the city of Amsterdam. The URL for the video page is

<http://www.cs.vu.nl/video/index-en.html>

✓ This URL consists of three parts: the protocol (*http*), the DNS name of the host (*www.cs.vu.nl*), and the file name (*video/index-en.html*), with certain punctuation separating the pieces. The file name is a path relative to the default Web directory at *cs.vu.nl*.

HTTP Overview

- ✓ The protocol used for communication between a **browser** and a **web server** or between intermediate machines and web servers is known as Hyper Text Transfer protocol (HTTP).
- ✓ HTTP defines how Web clients request Web pages from Web servers and how servers transfer Web pages to clients. The general idea is illustrated in Figure below.



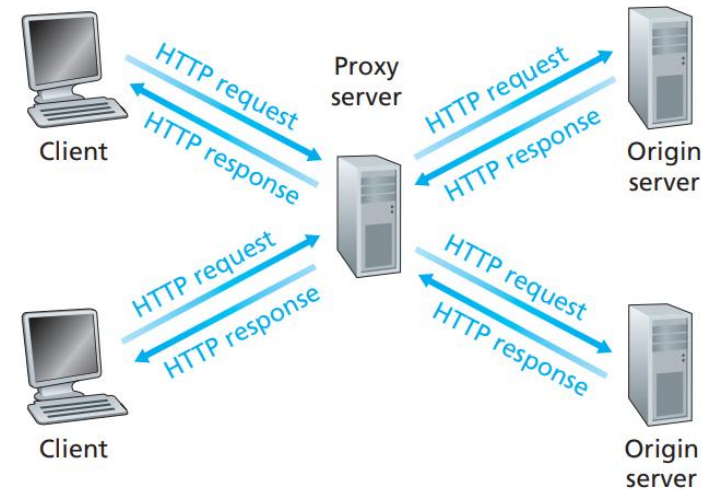
HTTP request-response behavior

- ✓ **Web browsers** (such as Internet Explorer and Firefox) implement the client side of HTTP, we will use the words browser and client interchangeably. **Web servers**, which implement the server side of HTTP addressable by a URL.
- ✓ The HTTP client first initiates a TCP connection with the server. Once the connection is established, the browser and the server processes access TCP through their socket interfaces.
- ✓ On the client side the socket interface is the door between the client process and the TCP connection; on the server side it is the door between the server process and the TCP connection.

- ✓ Once the TCP connection is in place, the browser and web server use HTTP to communicate; the browser sends a request to retrieve a specific page, and the server responds by sending a copy of the page.
- ✓ In most cases, a browser requests a Web page, and the server transfers a copy to the browser. HTTP also allows transfers from browser to server (form upload).

Web Caching

❖ A **Web cache** also called a **proxy server** is a network entity that satisfies HTTP requests on the behalf of an origin **Web server**. The Web cache has its own disk storage and keeps copies of **recently requested objects** in this storage.

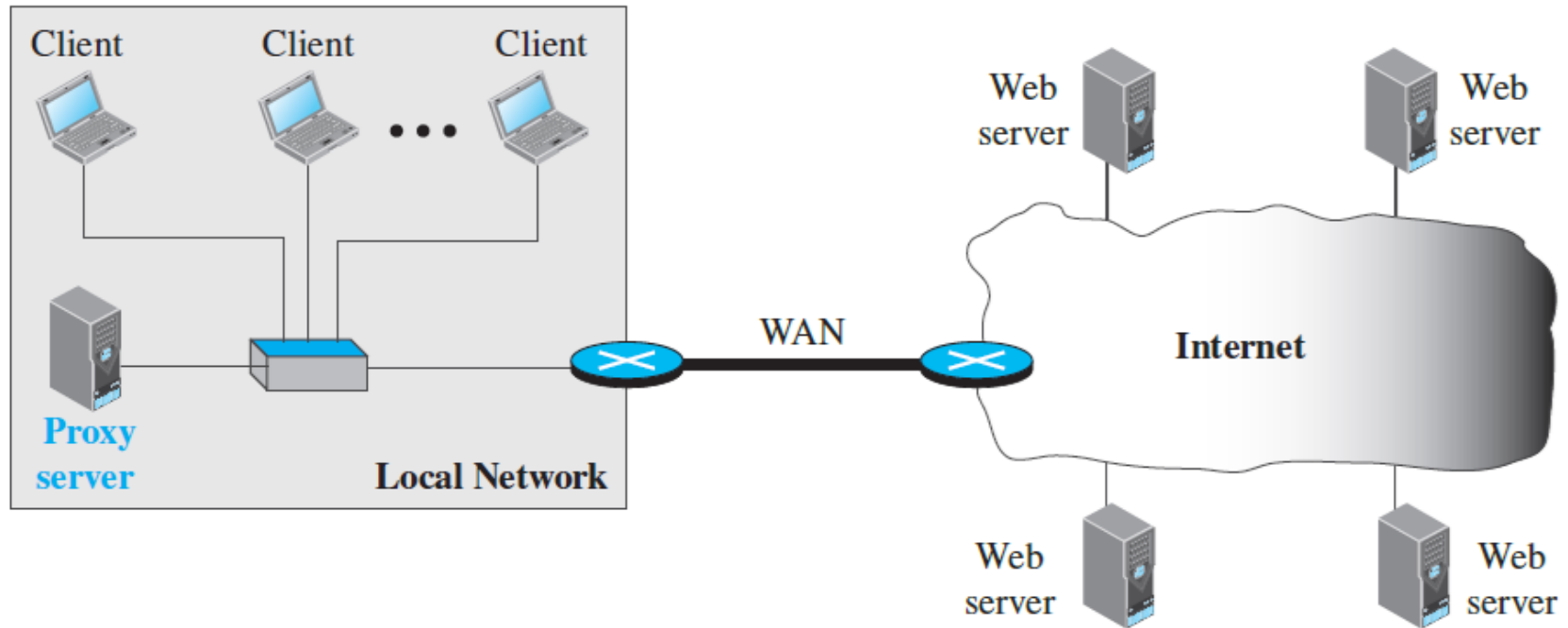


Proxy Server Location

The proxy servers are normally located at the client site. This means that we can have a hierarchy of proxy servers:

- ✓ A client computer can also be used as a proxy server, in a small capacity, that stores responses to requests often invoked by the client.
- ✓ In a company, a proxy server may be installed on the computer LAN to reduce the load going out of and coming into the LAN.
- ✓ An ISP with many customers can install a proxy server to reduce the load going out of and coming into the ISP network.

Figure below shows an example of a use of a proxy server in a local network, such as the network on a campus or in a company.



Q. What are the possible locations of proxy server?

The caching process works as follows:

1. The browser establishes a TCP connection to the Web cache and sends an HTTP request for the object to the Web cache
2. The Web cache checks to see if it has a copy of the object stored locally called **cache hit**. If it does, the Web cache returns the object within an HTTP response message to the client browser.
3. If the resource is not cached, **cache miss** will result. Web cache opens a TCP connection to the origin server and then sends an HTTP request for the object into the original server. After receiving this request, the origin server sends the object within an HTTP response to the Web cache.
4. When the Web cache receives the object, it stores a copy in its local storage and sends a copy, within an HTTP response message, to the client browser

Caching provides four benefits:

- a. Reduce load on the web server for example during admission test of Jahangir University.
- b. Reduce latency
- c. Reduce bandwidth utilization of users
- d. Network traffic reduction

Q. Steps of web caching process and its benefits.

Telnet and Secure Shell

- ✓ Telnet (teletype network) is a remote login protocol uses TCP/IP standard for establishing connection to a remote system. Recently IT specialists realize that telnet traffic is plain text hence suffers from security.
- ✓ To combat the situation, a new protocol SSH (or Secure Shell) is used to log into a remote machine which ensure the transmission of data thorough encrypted channel. We can use the SSH server to enable an SSH client for **encrypted communications** with Cisco port. Because of security the telnet protocol is replaced by Secure Shell (SSH) protocol.

File Transfer Protocol (FTP)

- ✓ File transfer is an important computer networking application. It is always essential that files and information geographically distributed over different location be shared among the numbers of working group.
- ✓ File Transfer Protocol (FTP) is part of the TCP/IP suite and is very similar to TELNET. Both FTP and TELNET are built on the client server paradigm, and both allow a user to establish a remote connection. However, TELNET provides a broader access to a user, whereas FTP allows access only to certain files. The sequence of operation of FTP is like:

1. A user requests a connection to a remote server
 2. The user waits for an acknowledgement.
 3. Once connected, the user must enter a user ID, followed by password.
 4. The connection is established over TCP session.
 5. The desired files are transferred.
 6. The user closes the FTP connection.
- FTP can also run through a Web browser.

Well-Known TCP Port Numbers

Ports are numbered and used as global standards to identify specific processes or types of network services provided at transport layer imposed from application layer. Ports indicate which service is desired. For example, TCP port 25 is for mail, and TCP port 80 is for HTTP.

20	FTP -- Data
21	FTP -- Control
22	SSH Remote Login Protocol
23	Telnet
25	SMTP
53	DNS
80	HTTP
110	POP3
161	SNMP

Network Management

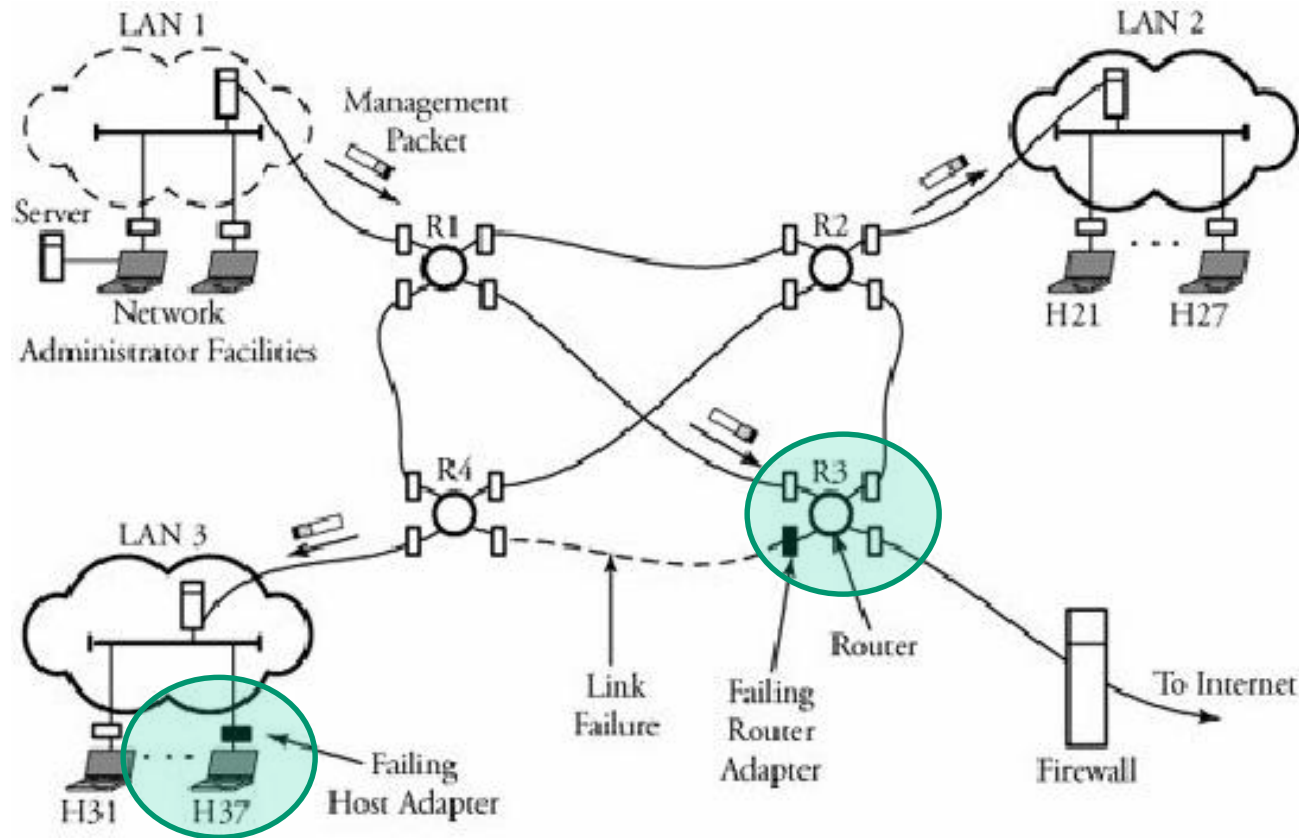
- ✓ The main purpose of network management is to monitor, manage, and control a network. A network can be structured with many **links, routers, switches, servers, and other physical-layer devices**, which can be equipped with many network protocols that coordinate them.
- ✓ Imagine when thousands of such devices or protocols are tied together by an ISP/organization network and how drastic their management can become to avoid any interruptions in routine services.
- ✓ In this context the purpose of **network management** is to monitor, test, and analyze the hardware and software of components of the entire network.

Q. Why network management software is necessary?

Network management tasks can be characterized as follows:

QoS and performance management. A network administrator periodically monitors and analyzes routers, hosts, and utilization of links and then redirect traffic flow (in case of congestion) to avoid any overloaded spots. Certain tools are available to detect rapid changes in traffic flow.

Network failure management. Any fault in a network, such as link, host, or router hardware or software outages, must be detected, located, and responded to by the network.



Simple network management in a scenario of LANs connecting to the Internet₄₂

Security management. A network administrator is responsible for the security of its network. This task is handled mainly through firewalls. A **firewall** can monitor and control access points.

Billing and accounting management. The network administrator specifies user access or restrictions to network resources and issues all billing and charges, if any, to users.

BW management, for example using **MikroTik router**.

Q. Mention few tasks of network management software.

Few Network Management Tools:

Auvik, Datadog Network Monitoring, PRTG Network Monitor Nagios XI, and ManageEngine OpManager, Azure Virtual WAN, Cisco FindIT Network Management